CITS3002: Lab 1 (Solutions)

Exercise 1:

1-1: Check your IP configuration (run "`ipconfig/all`" for Windows or "`ifconfig`" for Mac/Linux in the terminal).

Answer:

By running "`ipconfig/all`" in Windows, you will get an output similar to the following:

```
 Connection-specific DNS Suffix  . : guest.uwa.edu.au
Description . . . . . . . . . . . : Intel(R) Wi-Fi 6E 160MHz
Physical Address. . . . . . . . . : E8-B8-4A-9C-8A-B2
DHCP Enabled. . . . . . . . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
IPv4 Address. . . . . . . . . . . : 10.143.120.176
Subnet Mask . . . . . . . . . . . : 255.255.128.0
Lease Obtained. . . . . . . . . . : Tuesday, February 24, 2026 10:08:50 AM
Lease Expires . . . . . . . . . . : Tuesday, February 24, 2026 8:08:50 PM
Default Gateway . . . . . . . . . : 10.140.0.1
DHCP Server . . . . . . . . . . . : 10.37.10.230
DHCPv6 IAID . . . . . . . . . . . : 167294285
DHCPv6 Client DUID. . . . . . . . : 00-01-01-00-30-B5-35-79-F8-B5-4D-9D-7F-D3
DNS Servers . . . . . . . . . . . : 10.36.13.180
```

1-2: What is your IP address?

Answer: 10.143.120.176

1-3: What is your MAC (physical) address?

Answer: E8-B8-4A-9C-8A-B2

1-4: Explain which layer of the OSI model each of these addresses belongs to.

Answer: The IP address 10.143.120.176 is a logical address assigned to your device at the Network Layer (Layer 3) of the OSI model. It identifies your device on the network. The MAC address E8-B8-4A-9C-8A-B2 is a physical (hardware) address assigned to your device's network interface card (NIC). It operates at the Data Link Layer (Layer 2) of the OSI model.

Explain why each of these addresses (MAC and IP) is classified as either physical or logical.

Answer: The IPv4 address is considered logical because it is not permanently tied to hardware. It represents the device's location in a network structure rather than the physical device itself and it can change when you connect to a different network. The MAC address is considered physical because it is associated with the hardware of the network interface card. It uniquely identifies the device at the data link layer and is typically assigned by the manufacturer.

Exercise 2:

Compare Bus, Star, and Mesh topologies:

2-1: Write at least one advantage of each topology.

Answer:

Bus topology is simple and inexpensive because it requires minimal cabling and no central device.

Star topology is easy to manage and expand since each device connects to a central hub or switch.

Mesh topology offers high reliability and fault tolerance because multiple paths exist between devices.

2-2: Write at least one disadvantage of each topology.

Answer:

Data transmitted on a bus topology is visible to all connected devices. A failure in the backbone cable can bring down the entire network.

Star topology has a single point of failure at the central hub or switch, and it requires more cabling than bus topology, which increases installation costs.

Mesh topology is expensive due to the large amount of cabling and hardware required, and it is complex to configure and maintain, especially as the network grows.

2-3: Which topology is most commonly used today, and why? Give an example?

Answer:

Star topology is the most commonly used today because it is easy to manage and scalable. It allows devices to connect through a central switch or router, making troubleshooting and expansion straightforward. A common example is a home Wi-Fi network, where all devices connect to a central wireless router.

Exercise 3:

For each of the protocols HTTP, TCP, and IP, identify:

3-1: Which layer does it belong to in the OSI model? What is the main functionality of each protocol?

Answer: In the OSI model, HTTP belongs to the application layer (layer 7), TCP belongs to the transport layer (layer 4), and IP belongs to the network layer (layer 3). HTTP operates at the application level to enable communication between web clients and servers, TCP provides reliable end-to-end data transmission at the transport layer, and IP is responsible for logical addressing and routing of packets across networks at the network layer.

3-2: Which layer does it belong to in the TCP/IP model?

Answer: In the TCP/IP model, HTTP belongs to the application layer, TCP belongs to the transport layer, and IP belongs to the internet layer.

3-3: What is the primary function of the corresponding layer in the OSI model for each protocol?

Answer: The primary function of the application layer (where HTTP belongs) is to provide network services directly to end-user applications, enabling communication such as web browsing and email. The primary function of the transport layer (where TCP belongs) is to ensure reliable end-to-end communication through mechanisms such as congestion control. The primary function of the network layer (where IP belongs) is to provide routing, allowing data packets to travel across multiple interconnected networks from source to destination.